

High School Physics – Study Material

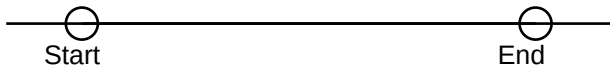
Chapter 1: Motion

Motion is the change in position of an object with respect to time.
It is described using distance, displacement, speed, and velocity.

Distance is the total path covered by an object.

Displacement is the shortest distance between initial and final positions.

Diagram: Object moving in a straight line



Chapter 2: Newton's Laws of Motion

Newton's First Law (Law of Inertia):

An object remains at rest or in uniform motion unless acted upon by an external force.

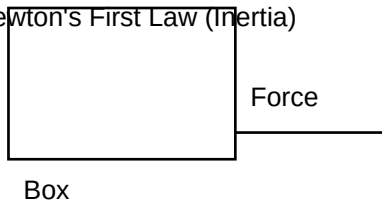
Newton's Second Law:

Force is equal to the rate of change of momentum ($F = ma$).

Newton's Third Law:

For every action, there is an equal and opposite reaction.

Diagram: Newton's First Law (Inertia)



Chapter 3: Universal Law of Gravitation

Every object in the universe attracts every other object with a force.
This force depends on the masses of the objects and the distance between them.

Gravitational Force Formula:

$$F = G \times (m_1 \times m_2) / d^2$$

This law explains the motion of planets and objects falling on Earth.

Chapter 4: Work, Energy, and Power

Work is said to be done when a force causes displacement.

Energy is the capacity to do work.

Power is the rate of doing work.

Examples:

- Lifting a book requires work.
- A moving car has kinetic energy.
- Electric bulbs consume power.